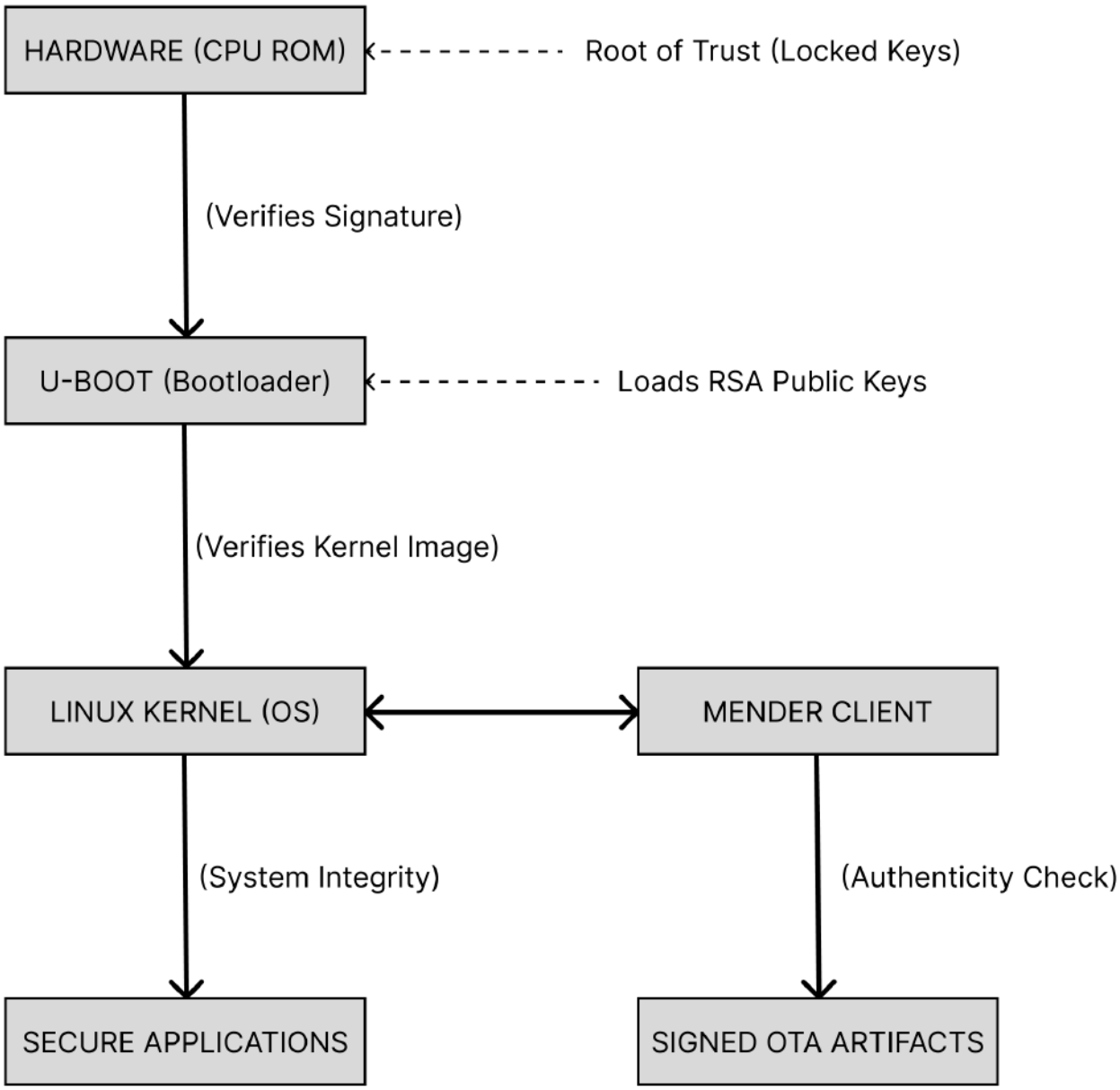
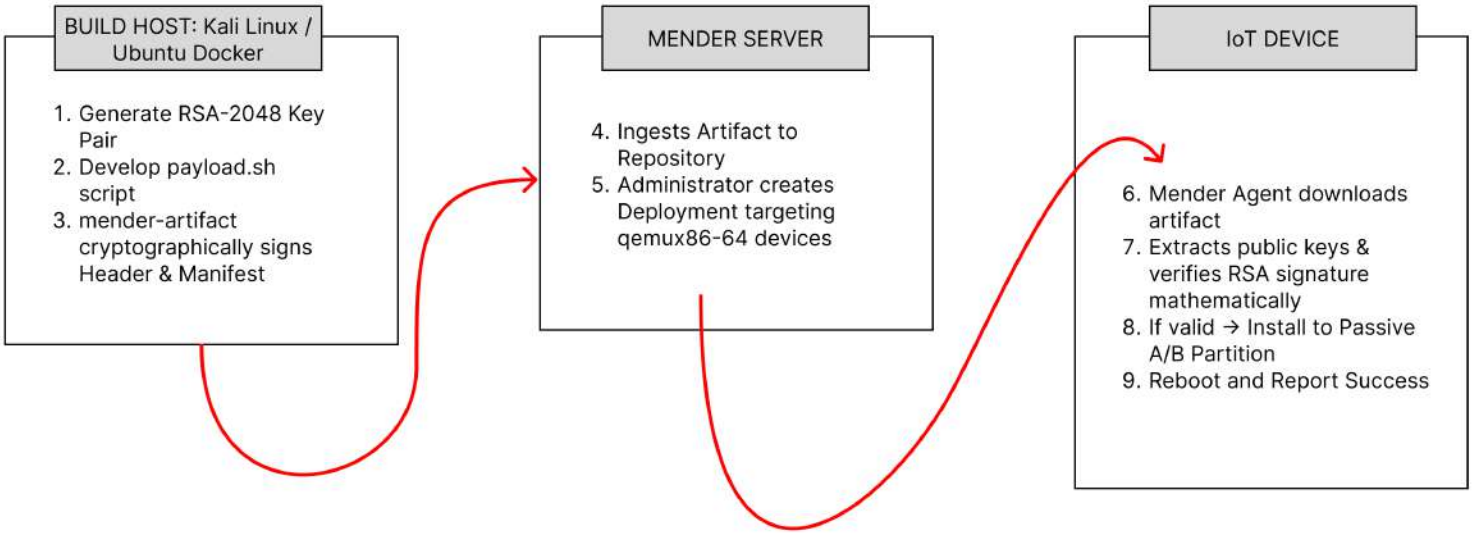
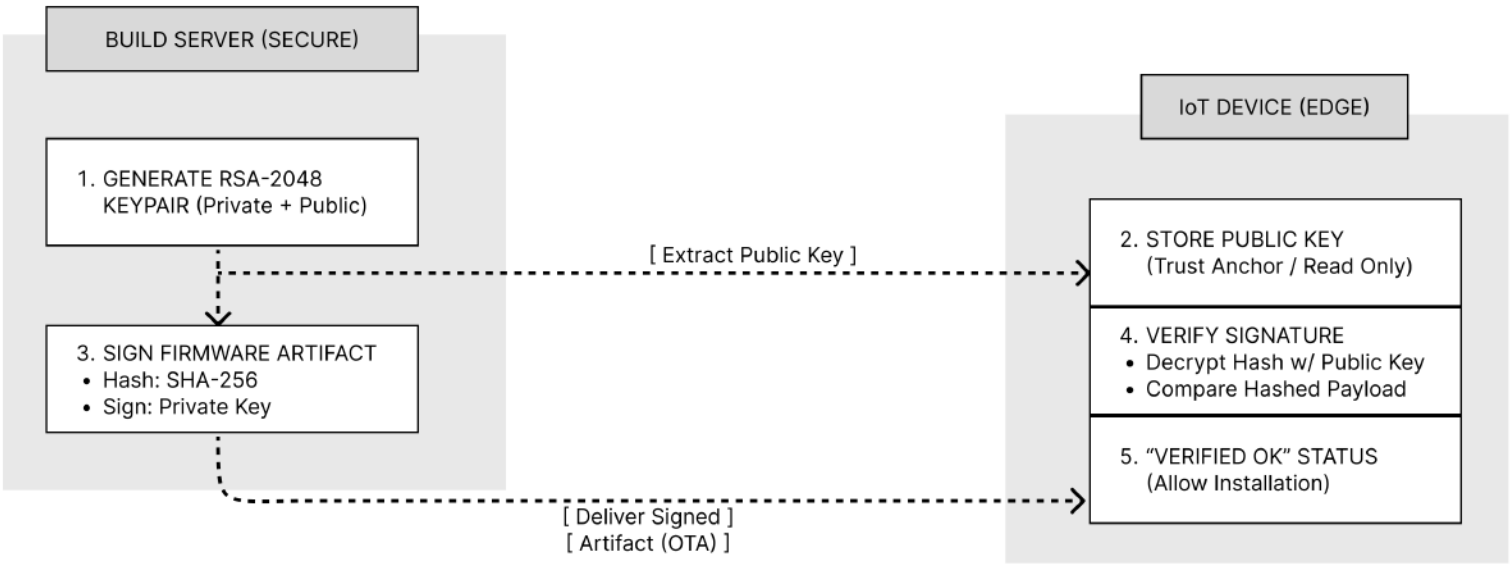
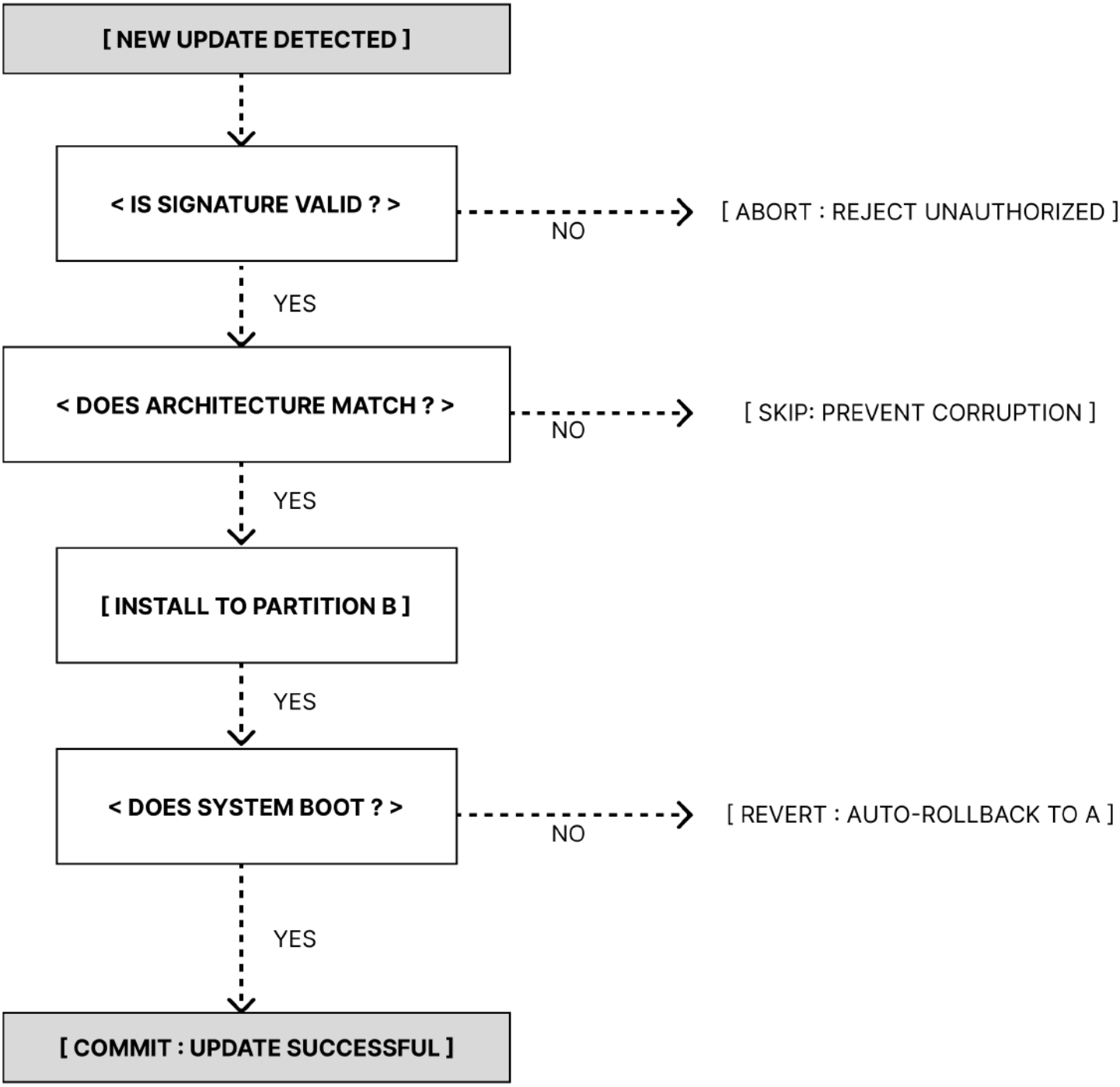
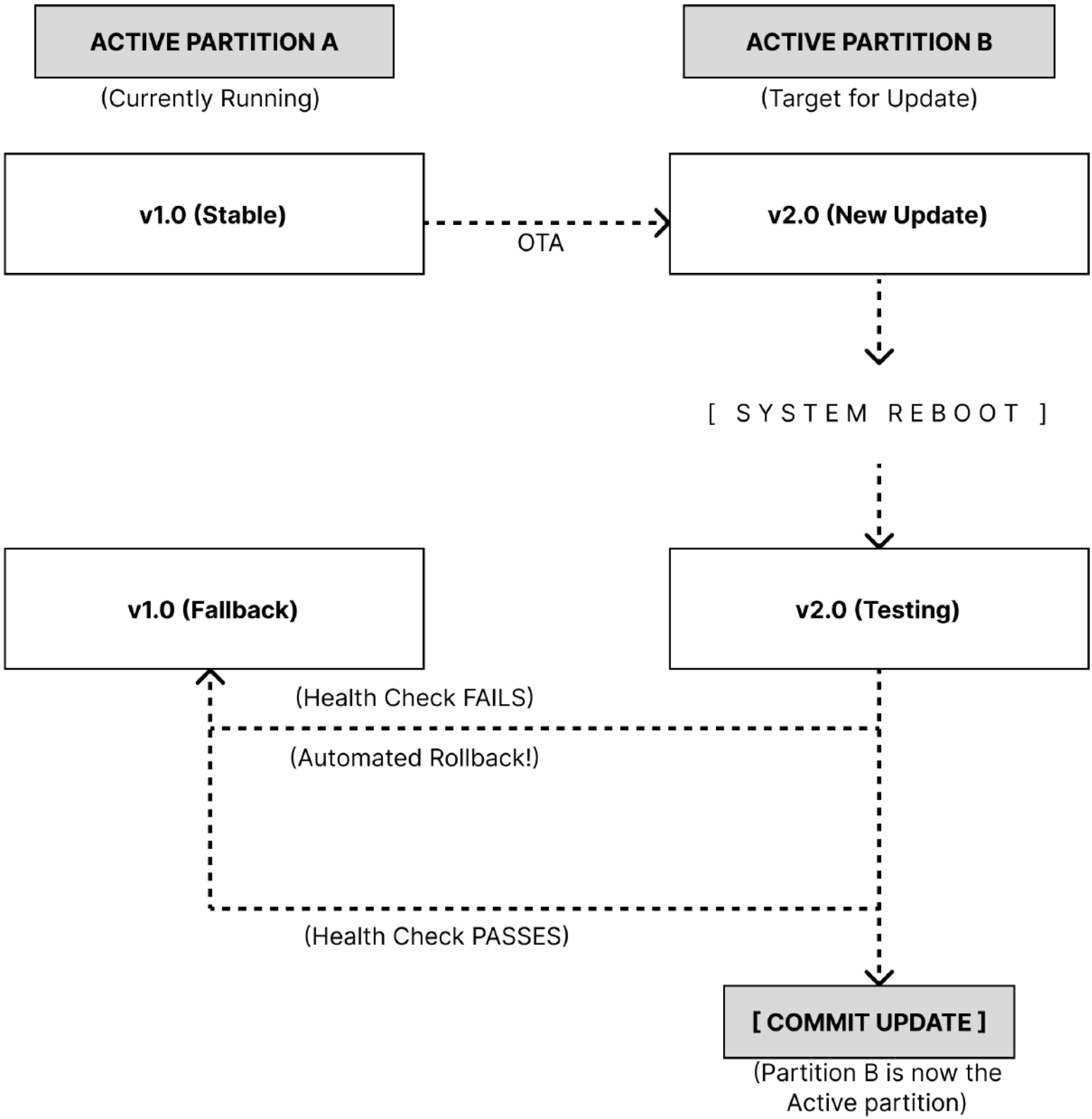
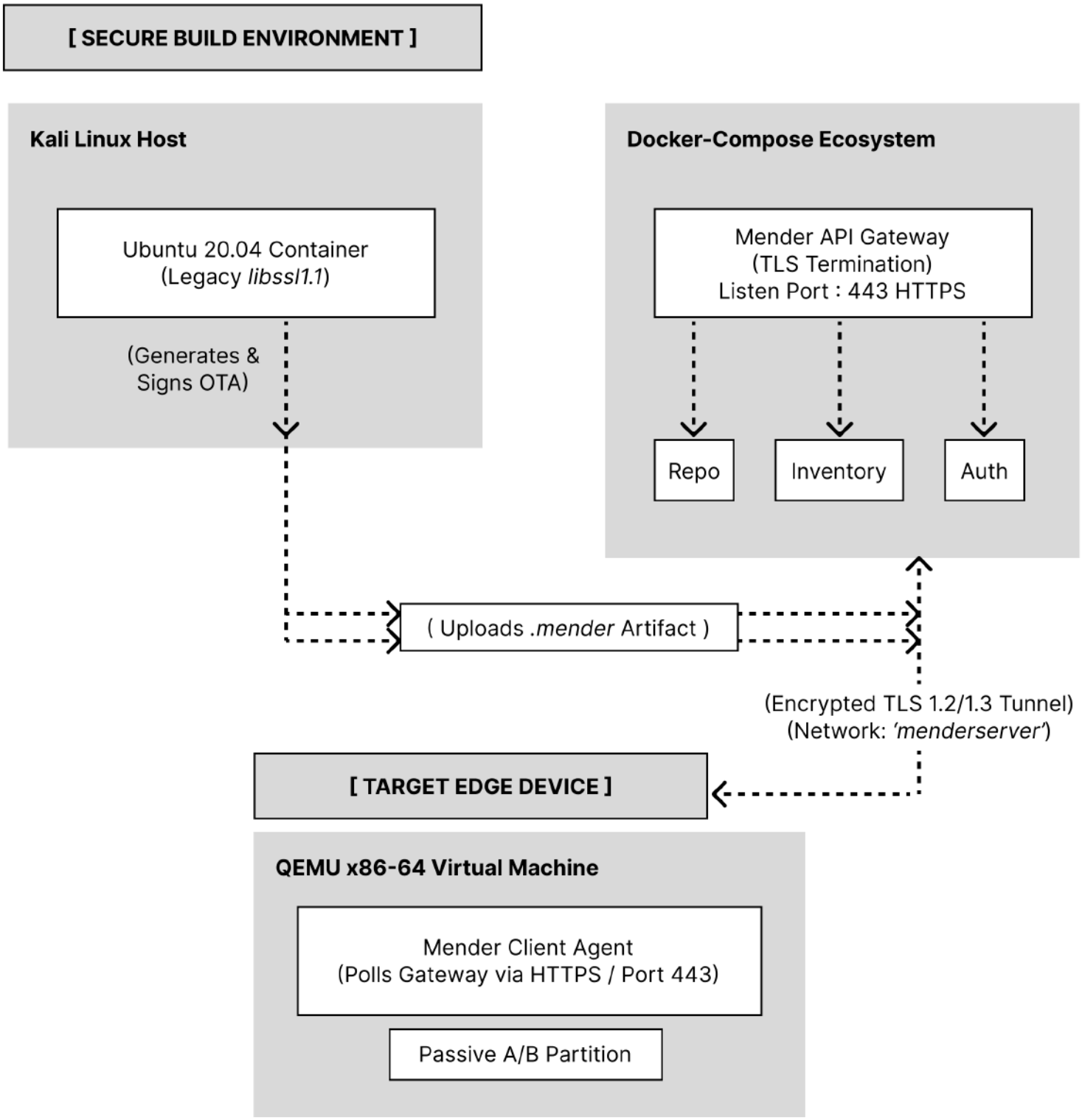

HARDWARE (CPU ROM)
Root of Trust (Locked Keys)
(Verifies Signature)
U-BOOT (Bootloader)
Loads RSA Public Keys
(Verifies Kernel Image)
LINUX KERNEL (OS)
MENDER CLIENT
(System Integrity)
(Authenticity Check)
SECURE APPLICATIONS
SIGNED OTA ARTIFACTS

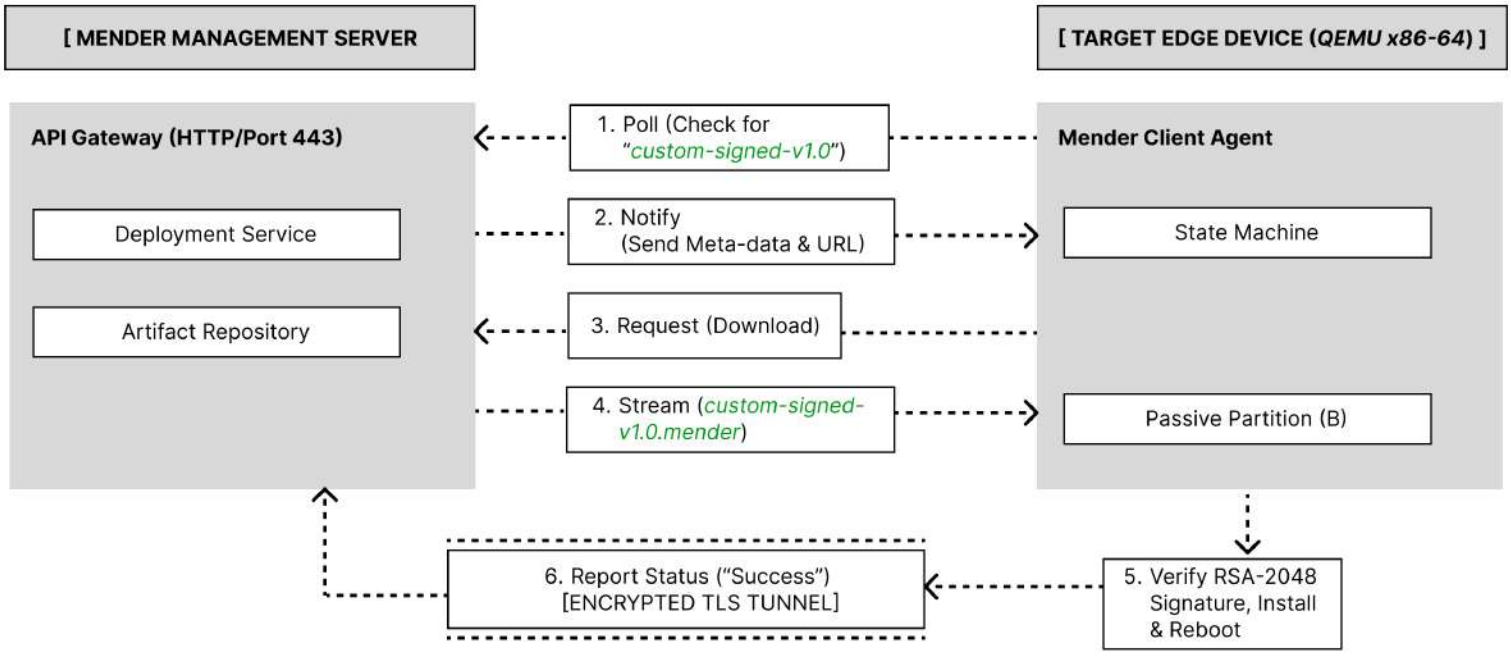

BUILD HOST: Kali Linux / Ubuntu Docker
1. Generate RSA-2048 Key Pair
2. Develop payload.sh script
3. mender-artifact cryptographically signs Header & Manifest
MENDER SERVER
4. Ingests Artifact to Repository
5. Administrator creates Deployment targeting qemux86-64 devices
IoT DEVICE
6. Mender Agent downloads artifact
7. Extracts public keys & verifies RSA signature mathematically
8. If valid → Install to Passive A/B Partition
9. Reboot and Report Success

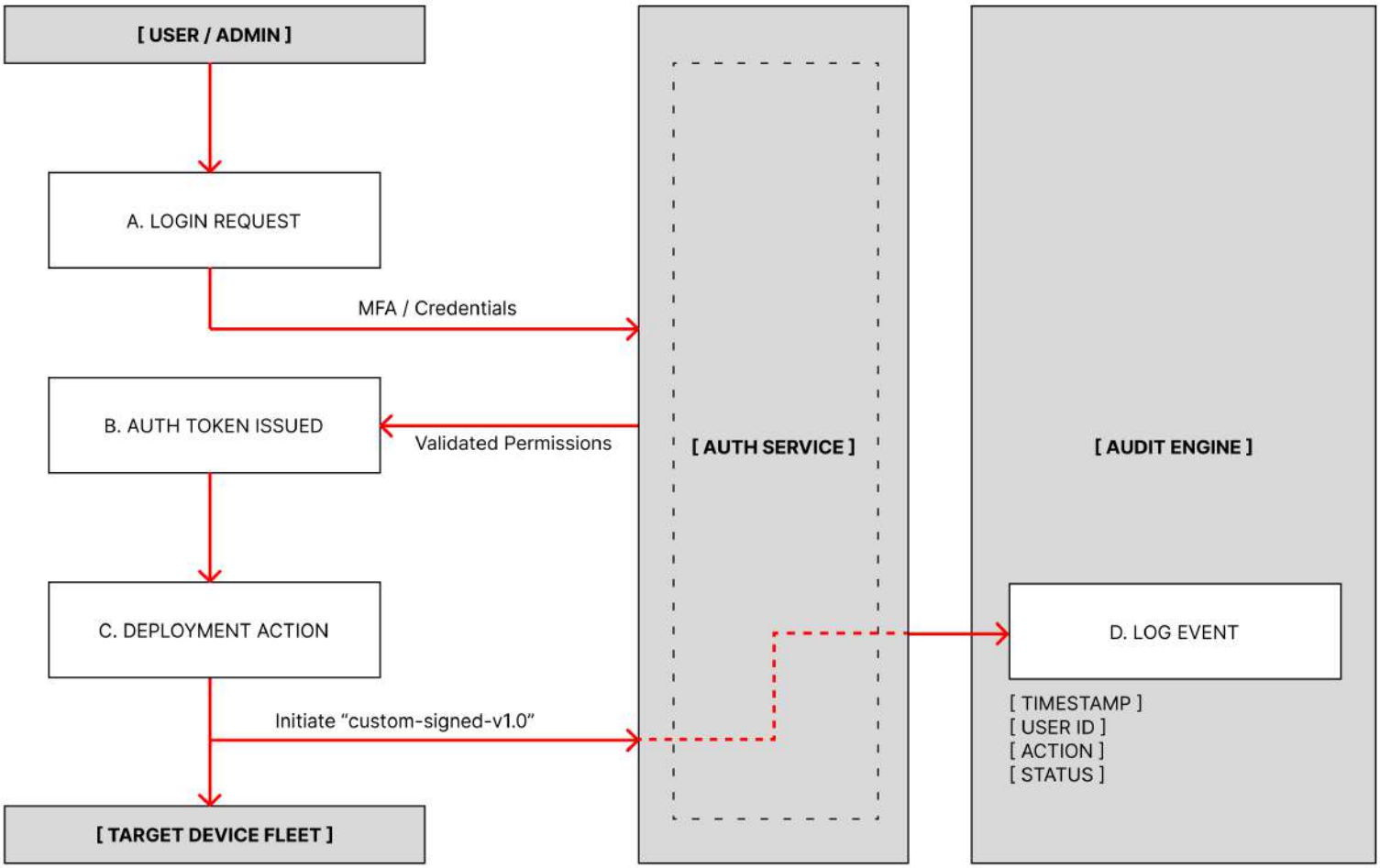

BUILD SERVER (SECURE)
1. GENERATE RSA-2048 KEYPAIR (Private + Public)
3. SIGN FIRMWARE ARTIFACT
• Hash: SHA-256
• Sign: Private Key
[ Extract Public Key ]
[ Deliver Signed ] [ Artifact (OTA) ]
IoT DEVICE (EDGE)
2. STORE PUBLIC KEY (Trust Anchor / Read Only)
4. VERIFY SIGNATURE
• Decrypt Hash w/ Public Key
• Compare Hashed Payload
5. "VERIFIED OK" STATUS (Allow Installation)

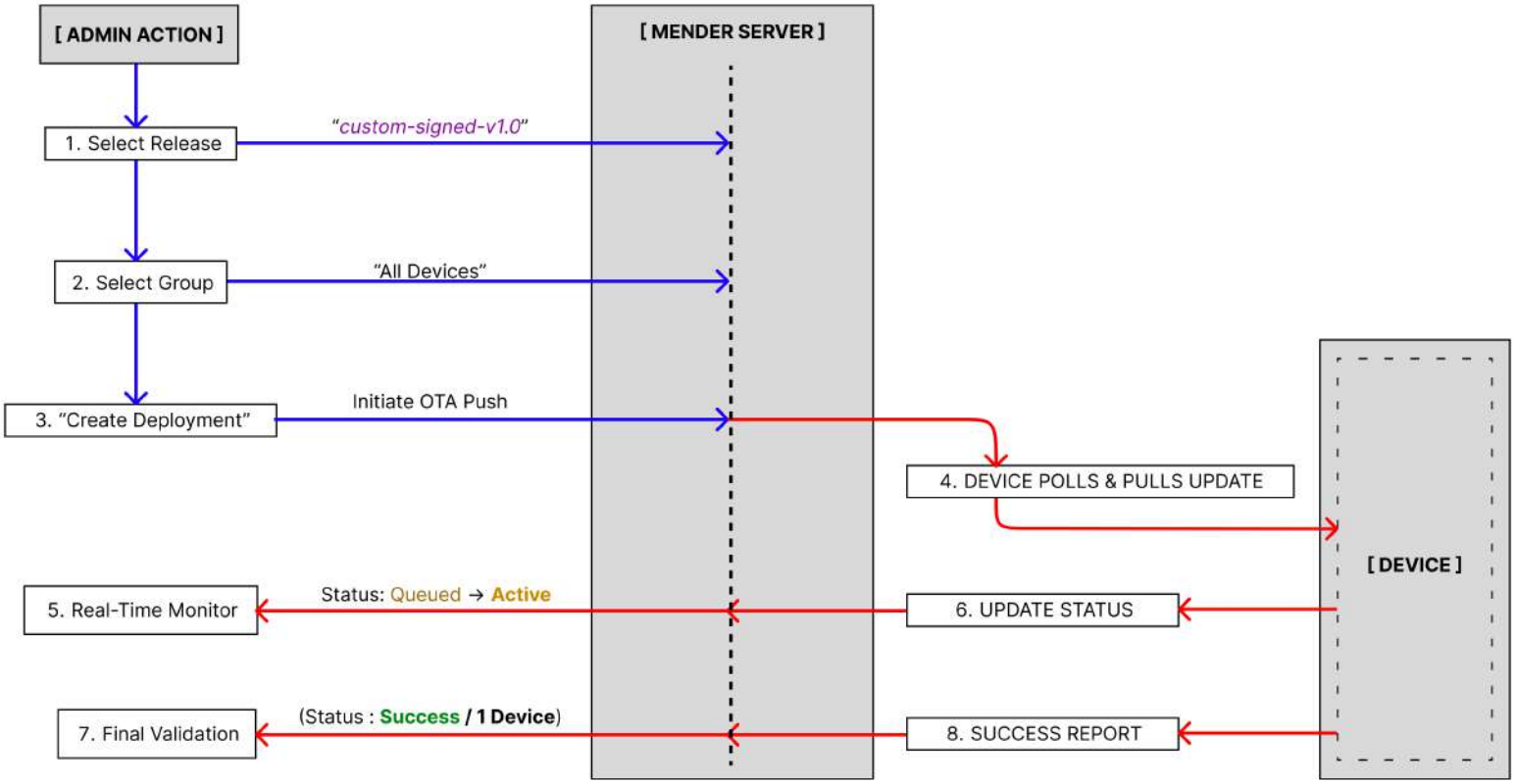

[ NEW UPDATE DETECTED ]
< IS SIGNATURE VALID ? >
NO
[ ABORT : REJECT UNAUTHORIZED ]
YES
< DOES ARCHITECTURE MATCH ? >
NO
[ SKIP: PREVENT CORRUPTION ]
YES
[ INSTALL TO PARTITION B ]
YES
< DOES SYSTEM BOOT ? >
NO
[ REVERT : AUTO-ROLLBACK TO A ]
YES
[ COMMIT : UPDATE SUCCESSFUL ]

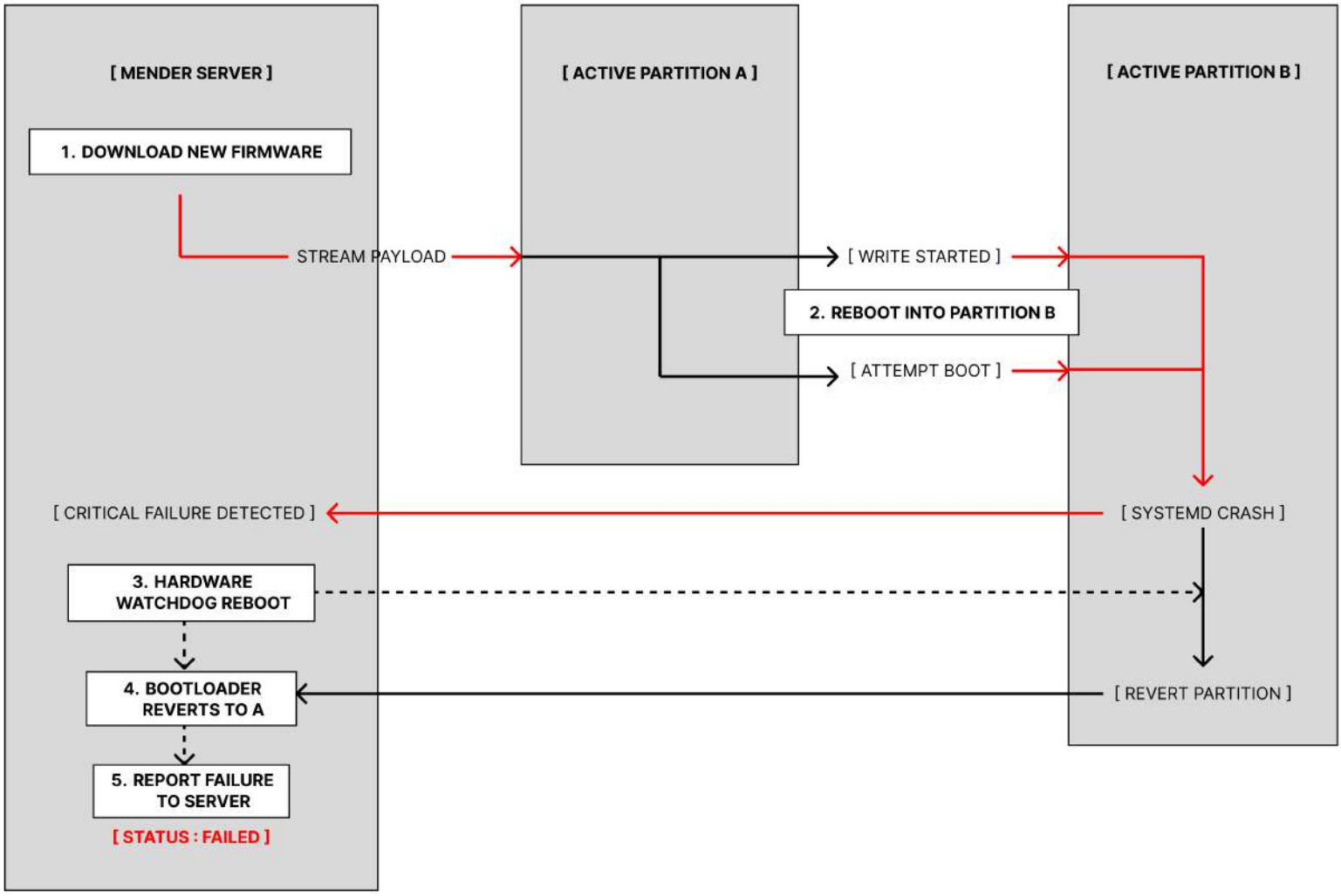

ACTIVE PARTITION A
(Currently Running)
ACTIVE PARTITION B
(Target for Update)
v1.0 (Stable)
OTA
v2.0 (New Update)
[ SYSTEM REBOOT ]
v1.0 (Fallback)
v2.0 (Testing)
(Health Check FAILS)
(Automated Rollback!)
(Health Check PASSES)
[ COMMIT UPDATE ]
(Partition B is now the Active partition)


[ SECURE BUILD ENVIRONMENT ]
Kali Linux Host
Ubuntu 20.04 Container
(Legacy libssl1.1)
(Generates &
Signs OTA)
Docker-Compose Ecosystem
Mender API Gateway
(TLS Termination)
Listen Port : 443 HTTPS
Repo
Inventory
Auth
( Uploads .mender Artifact )
(Encrypted TLS 1.2/1.3 Tunnel)
(Network: 'menderserver')
[ TARGET EDGE DEVICE ]
QEMU x86-64 Virtual Machine
Mender Client Agent
(Polls Gateway via HTTPS / Port 443)
Passive A/B Partition


[ MENDER MANAGEMENT SERVER
[ TARGET EDGE DEVICE (QEMU x86-64) ]
API Gateway (HTTP/Port 443)
Deployment Service
Artifact Repository
1. Poll (Check for "custom-signed-v1.0")
2. Notify (Send Meta-data & URL)
3. Request (Download)
4. Stream (custom-signed-v1.0.mender)
Mender Client Agent
State Machine
Passive Partition (B)
5. Verify RSA-2048 Signature, Install & Reboot
6. Report Status ("Success") [ENCRYPTED TLS TUNNEL]


[ USER / ADMIN ]
A. LOGIN REQUEST
MFA / Credentials
B. AUTH TOKEN ISSUED
Validated Permissions
C. DEPLOYMENT ACTION
Initiate “custom-signed-v1.0”
[ TARGET DEVICE FLEET ]
[ AUTH SERVICE ]
[ AUDIT ENGINE ]
D. LOG EVENT
[ TIMESTAMP ]
[ USER ID ]
[ ACTION ]
[ STATUS ]


[ ADMIN ACTION ]
[ MENDER SERVER ]
1. Select Release
"custom-signed-v1.0"
2. Select Group
"All Devices"
3. "Create Deployment"
Initiate OTA Push
4. DEVICE POLLS & PULLS UPDATE
[ DEVICE ]
5. Real-Time Monitor
Status: Queued → Active
6. UPDATE STATUS
7. Final Validation
(Status : Success / 1 Device)
8. SUCCESS REPORT


[ MENDER SERVER ]
1. DOWNLOAD NEW FIRMWARE
STREAM PAYLOAD
[ ACTIVE PARTITION A ]
[ WRITE STARTED ]
2. REBOOT INTO PARTITION B
[ ATTEMPT BOOT ]
[ ACTIVE PARTITION B ]
[ CRITICAL FAILURE DETECTED ]
[ SYSTEMD CRASH ]
3. HARDWARE WATCHDOG REBOOT
4. BOOTLOADER REVERTS TO A
[ REVERT PARTITION ]
5. REPORT FAILURE TO SERVER
[ STATUS : FAILED ]